

INSTITUTE LEVEL SKILL COMPETITION—2026
GOVT. POLYTECHNIC HAMIRPUR
WEB APPLICATION DEVELOPMENT

Duration: 2 Hours

Maximum Marks: 100

PROBLEM STATEMENT — “Student Technical Workshop Registration System”

Develop a functional web application for managing registration for technical workshops organized by the institute. Students should be able to view available workshops and register online. An administrator should be able to view registered students and workshop-wise registration counts.

1. TECHNOLOGY

Students may use any suitable web technology/framework. Suggested: HTML5, CSS3, Javascript, PHP + MySQL OR Python Flask + SQLite/MySQL. Bootstrap or other suitable frameworks may be used.

2. REQUIRED FUNCTIONALITY

A. Home / Workshop Page

- Display institute name, Technical Workshop 2026, date, venue, brief introduction and navigation menu.
- Display available workshops with a Register Now option.
- Sample workshops: Python Programming, Web Development, IoT with Arduino, Database Management.
- For each workshop display: Workshop ID | Name | Date | Time | Venue | Seats Available | Description.
- Workshop information **MUST** be retrieved from the database.

B. Student Registration

Field	Requirement
Student Name	Required
Roll Number	Required
Branch	Required
Semester	Required
Email	Required
Mobile Number	Required
Workshop	Required
Gender	Appropriate input
Declaration	Required
Submit / Reset	Required

C. Validation

- Implement client-side and server-side validation.
- Check required fields, valid email/mobile/roll number, valid workshop selection and declaration acceptance.
- Invalid/unknown workshop must not be accepted.
- A student must not be allowed to register more than once for the same workshop using the same roll number.

D. Database

Create at least two related tables:

Workshops: workshop_id, workshop_name, workshop_date, workshop_time, venue, max_seats, description

Registrations: registration_id, student_name, roll_no, branch, semester, email, mobile, gender, workshop_id, registration_date

Use an appropriate relationship between Workshops and Registrations.

3. REGISTRATION PROCESSING

- Receive submitted data at the backend and validate it.
- Store valid registration in the database.
- Display a clear confirmation message with registration details.
- The system must not allow registration when the workshop has reached its maximum participant limit.

4. ADMIN / REGISTRATION VIEW

Create a simple administrator page showing the following registration summary:

Workshop	Registered
Python Programming	12
Web Development	15
IoT with Arduino	10
Database Management	8
Total	45

Display registered students in a table containing at least:

Registration ID | Student Name | Roll No. | Branch | Semester | Workshop

The workshop name should be obtained through the database relationship.

5. USER INTERFACE

- Clean and attractive layout with proper navigation, forms, controls, buttons and messages.
- Use CSS for styling and preferably make the application responsive.
- The application should preferably work on Desktop, Tablet and Mobile.

6. BONUS FEATURES

- Search by student name/roll number; filter by workshop or branch.
- Display remaining seats and/or additional dashboard statistics.
- Edit/delete registrations or add/edit/delete workshops.
- Printable registration confirmation.
- Improved visual design, CSS animations or other useful features.

7. ASSESSMENT SCHEME

Component	Marks
Home & Workshop Listing	10
Registration Form & UI	15
Client/Server Validation	15
Database Design & Connectivity	15
Registration Processing	15
Seat Limit & Duplicate Checking	10
Admin Registration View & Summary	10
Responsive/Visual Design	5
Bonus Features	5
TOTAL	100